masked or corrupted sequence of tokens, and iteratively corrects subsets of them.

Inference. We extract *independent* samples from the model. For Transformers, during inference, we perform standard autoregression in both the next- and multi-token trained settings. We do this either with greedy decoding or with nucleus sampling (Finlayson et al., 2024).

3.1. Seed-conditioning

While temperature sampling is the standard way to elicit diversity from a Transformer, we speculate that this can lead to cognitive overload: the model must process multiple thoughts to compute a diverse softmax distribution. Alternatively, we propose conditioning on a random seed as input to the model, hoping that a thought can be fleshed out with singular focus. Thus, during training, each point is associated with an arbitrary (meaningless) prefix (e.g., of random characters) called a seed (note that there is no prefix in our tasks besides the seed). During test-time, novel seeds are used to extract test data. We can also view seed-conditioning more neutrally as a controlled way to simulate variations in prompt-wordings for a fixed task (e.g., `design a word problem` vs `construct a problem`). We elaborate on these intuitions in §5.1.

4. Experimental results

Key details. Part of our experiments are performed for a `Gemma v1 (2B)` pre-trained model (Gemma Team et al., 2024), averaged over 4 runs. For diffusion, we use a `90M` (non-embedding) parameters Score Entropy Discrete Diffusion model (`SEDD (90M)`; Lou et al., 2023). For a fair comparison against `NTP`, we use a `86M` (non-embedding) parameters `GPT-2 (86M)` model (Radford et al., 2019). In all our experiments, we finetune the models until it is clear that algorithmic creativity (Eq. 1) has saturated. All values are reported from this checkpoint. Finally, since our best Transformer results were under seed-conditioning (for both next- and mult-token training), our main results are reported under that training setting; we provide various ablations without that as well. Please see §D for more experimental details, and §C for precise dataset details (e.g., how the graph is constructed, how sequences are formatted etc.,).

4.1. Observations

Multi-token prediction improves algorithmic creativity significantly. In all our datasets, the algorithmic creativity of the `Gemma v1 (2B)` model increases significantly under multi-token prediction (Fig 3), with nearly a `5x` factor for the discovery datasets. Note that for this, we have selected the learning rate favorable towards next-token prediction; tuning for multi-token yields further gains (Fig 15). For

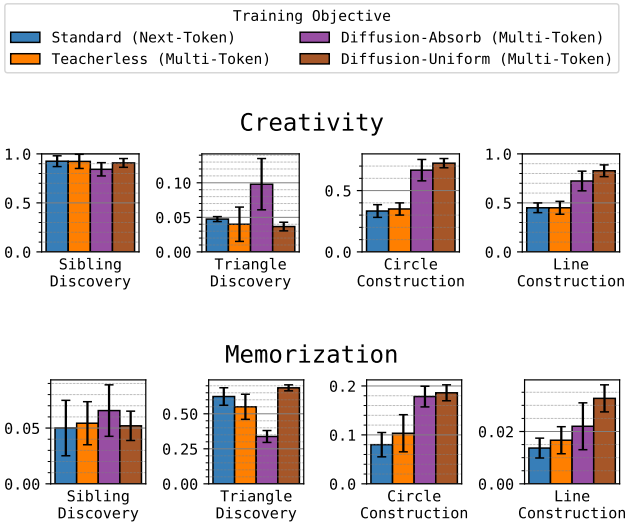

Figure 4. **Multi-token diffusion training improves algorithmic creativity (top; Eq 1) on three of our four open-ended tasks on `GPT-2 (86M)` and similarly-sized diffusion model, `SEDD (90M)`.** We report the best performance after tuning the sampling hyperparameters – temperature from $\{0, 0.5, 1, 2\}$, with or without top-K where $K = 50$.

the smaller models, in Fig 4, we find that the diffusion model improves over next-token training of similar-sized Transformers on all tasks, except on `Sibling Discovery` where it is mildly worse. With teacherless training, the gains are absent; however we find that the creativity of the top-K samples is generally improved (Fig 24), indicating a less peaky distribution. Regardless, the overall trend echoes prior findings that for smaller Transformers, teacherless training is hard to optimize and other multi-token objectives even hurt (Gloeckle et al., 2024) smaller models.

Multi-token prediction reduces memorization for the larger model. Algorithmic creativity may suffer either due to either incoherence or mode collapse or memorization. For our larger model, it is the last reason that dominates: across the board (in Fig 3), next-token prediction is significantly prone to memorizing the data, while the multi-token method is highly resistant. As foreshadowed in §2.6, we hypothesize that this is because `NTP` memorizes the earlier training tokens without a global plan, having fit the later tokens via local coherence rules (because of Clever Hans cheats à la B&N'24). In contrast, the smaller models have much lower absolute memorization values; the multi-token objective preserves or mildly worsens this, in a way that does not always hurt creativity proportionally. here, we find that amongst the top-K samples, teacherless training has reduced memorization for some tasks (see Fig 25).

We point the reader to §B.4 for further empirical evidence supporting our argument about `NTP` from §2.6, including

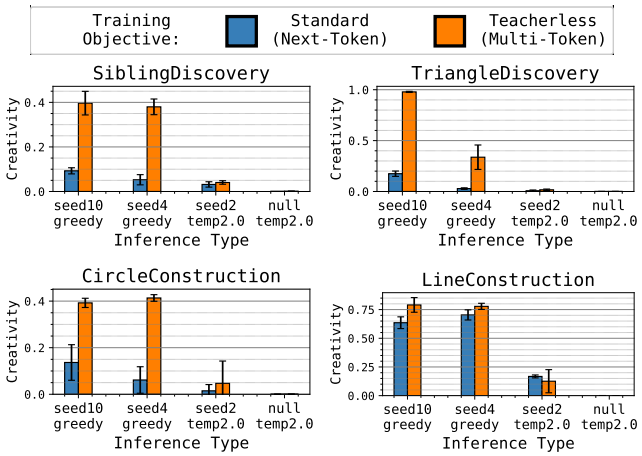

Figure 5. **Seed-conditioning significantly improves algorithmic creativity of both next- and multi-token prediction on `Gemma v1 (2B)` model.** The X-axis labels denote the prefix (at training and inference) and the temperature (at inference).

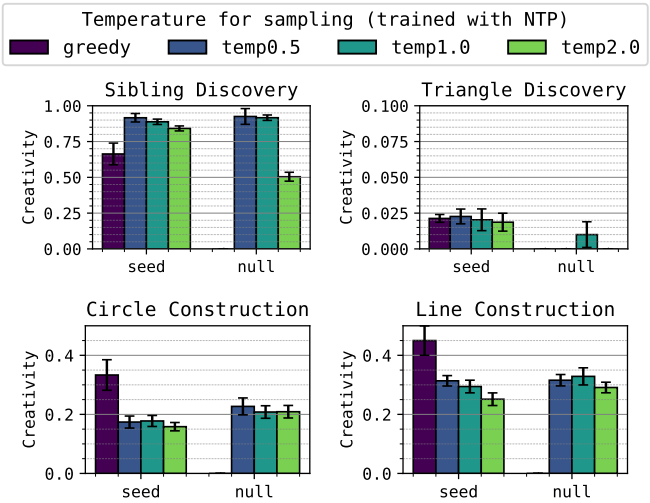

Figure 6. **Seed-conditioning achieves comparable algorithmic creativity in the `GPT-2 (86M)` model**: All models are trained with `NTP`. The color denotes temperature, while the X-axis, the prefix. Remarkably, seed-conditioning achieves comparable creativity even with greedy decoding, and improves creativity particularly in our exploratory creativity tasks (**bottom**).

experiments on token-reordering and experiments ruling out other hypotheses.

Seed-conditioning improves algorithmic creativity for Transformers. Orthogonal to the effect of multi-token vs. next-token objectives, we point out the effect of seed-conditioning a Transformer. First, and most remarkably, with seed-conditioning, there is no need for temperature: even deterministic greedy decoding generates diverse outputs and highly non-trivial creativity (Fig 5, Fig 6). This means that, even though the seeds we provide are arbitrarily paired with the training datapoint, the models have learned to transform the arbitrary noise into meaningful randomness. Next, increasing the seed lengths consistently boosts algorithmic creativity for both next-token and multi-token approaches (Fig 5, 23), presumably up to a limit. Thus, for Transformers, we propose viewing seed-conditioning as a knob for diversity distinct from temperature sampling.

Next, seed-conditioning results in algorithmic creativity comparable to temperature sampling. For the larger models (Fig 5), seed-conditioning generally outperforms the baseline "null" conditioning (which appears mode-collapsed, see Fig 18, 19, 20). For the smaller model as well (Fig 6), we find gains in all tasks except for `Sibling Discovery`. Besides, for any fixed temperature, prefixing a seed only improves performance over a null prefix (Fig 6, 18).

In §H.2, we find that the improved creativity from seed-conditioning comes from improved diversity, rather than from reduced memorization (unlike with multi-token training). All these behaviors of seed-conditioning require further study as they are non-trivial, especially given that the seeds are naively, arbitrarily chosen in relationship to the output. As a starting point, we provide some preliminary hypotheses for when and why seed-conditioning may help Transformers in §5.1. Note that we do not see gains of seed-conditioning when it comes to diffusion training (§ F.3).

Robustness to hyperparameters. In §E and Fig 22, we do sensitivity analysis on all the datasets. We report how our above findings are robust to the choice of learning rate, batch-size, number of training steps, weight given to the multi-token objective, varying sampling conditions and reasonable changes to the complexity of the dataset and training set size (as per our argument in §2.6, we do expect the next-vs. multi-token gap to diminish for larger dataset size).

4.2. An exploration of real-world summarization

For a more realistic examination, we finetune `GPT` models `NTP` and the multi-token teacherless objectives on summarization tasks (`XSUM`, `CNN/DailyMail`). We measure the diversity of a model for any given prompt by generating 5 different completions and computing a `Self-Bleu` metric (Zhu et al., 2018). Admittedly though, a summarization task is not as open-ended as we would like: a higher quality model (i.e., higher `Rouge`; Lin, 2004) necessarily means lower diversity. To account for this, we plot how diversity evolves over time as a function of generation quality; we then find in Fig 7 that for a given model quality, the larger multi-token models achieve higher diversity (albeit only by a slight amount). This increase does not hold for smaller models and is not always noticeable for `CNN/DailyMail` (see §I). Interestingly, teacherless training consistently shows an increase in summarization quality, measured by `Rouge`.

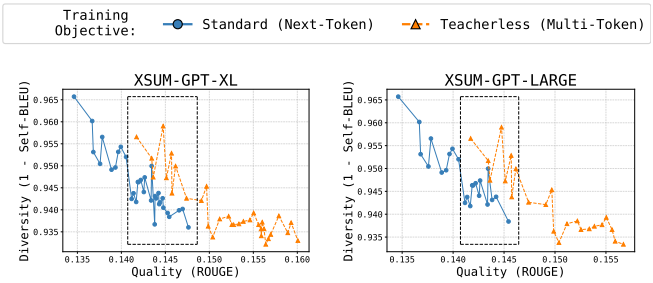

Figure 7. **Multi-token training improves diversity scores for `XSUM` summarization for large `GPT-2` models**: We observe gaps in diversity for a fixed quality over the course of finetuning.

5. Discussion

5.1. Intuition for seed-conditioning

Seed-conditioning produces non-trivial algorithmic creativity even with greedy decoding. Perhaps, one could view the seed prefixes as a simpler alternative to varying the wordings of a prompt (Li et al., 2023; Lau et al., 2024; Naik et al., 2024; Li et al., 2022) or tuning a soft-prompt (Wang et al., 2024c), both of which are known to induce diversity. However, it is unexpected that the seeds provide any meaningful randomness at all, given they are arbitrarily picked with no semantic relationship to the training point. We do not understand this presently.

Another pertinent question is to why seed-conditioning works better than temperature sampling in the settings it does. To this, we offer three potential explanations. The simplest explanation is that in the initial version of our paper, top-K sampling was enabled by default with $K = 50$ for the `GPT-2 (86M)` settings. This would have restricted the degree of freedom in the support induced by the output-layer randomness; thus, providing different seeds can help vary this support and produce more varying samples. However, even upon removing top-K sampling, we see gains in our exploratory creativity tasks for `GPT-2 (86M)` (after all their vocabulary is smaller than $K = 50$), and in all our `Gemma v1 (2B)` results. This is unexplained.

Thus, we speculatively put forth two more arguments. First, is a representational one. Fixing a random seed upfront may help the model flesh out (i.e., compute the tokens of) one thought per sample, as against maintaining a running set of multiple thoughts and computing a distribution over all their tokens at each step. A similar point is made in a concurrent position paper (Jahrens & Martinetz, 2025). The second argument is specific to next-token prediction on open-ended planning tasks: fixing a random seed upfront may help the model co-ordinate multiple interlocking random decisions in advance rather than deciding them on the fly.

Finally, there are also optimization aspects of how seed-conditioning works that we do not understand (see §B.1). Regardless, it remains to be seen whether seed-conditioning is useful in tasks beyond the minimal ones we design.

5.2. Effects of reasoning-enhancing methods.

Our argument is limited to learning open-ended tasks in a supervised manner which directly corresponds to how we perform pretraining or apply models for protein or drug discovery. While we do not comment on how well other powerful approaches like RL (DeepSeek-AI, 2025), chain-of-thought (CoT; Wei et al., 2022), and scaling test-time compute (OpenAI, 2024) would address the limitations we bring up, we make a few remarks. First, in cases where post-training only elicits pre-existing skills in the base model (Muennighoff et al., 2025), our limitations and suggestions about a next-token-trained base model remain relevant. Next, regardless of the benefits one may get from pre-training, our arguments imply that pre-training squanders its rich supervision, when it could be vastly compute- and data-efficient. Finally, there is a profound question as to whether exploration through sparse rewards can learn the creative process during training-time or whether articulation in the token space (with a thinking model) can efficiently maximize diversity during test-time (which may require brute-force enumeration of all candidates).

We present more discussions in §B.

6. Limitations

We enumerate in detail the limitations of our work in terms of our experimental conclusions and in terms of our general approach to an abstraction of creativity.

6.1. Limitations of our experimental conclusions

1. There may be many ways to improve upon next-token prediction for a minimal task. Unfortunately, success here does not necessarily guarantee success on more complex tasks. Conversely, minimal tasks are more valuable as a failure base case: failure here guarantees failure in more complex tasks.

2. Our examples do not preclude the existence of tasks where next-token prediction will outperform multi-token prediction; multi-token prediction is simply a more general-purpose objective suitable to lookahead tasks.

3. The teacherless multi-token prediction technique we explore as an alternative is generally harder to optimize than next-token prediction, especially for smaller models.

4. Even if multi-token approaches outperform next-token prediction relatively, in some of our simple tasks, all algorithms are far from delivering a sufficiently diverse